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Candidate Seat No. _____

GUJARAT UNIVERSITY

B. E. Semester -III (IT)

Data Structure and Algorithm

Date : 6th December, 2004]

[Time : 10.30A.M. to 1.30 P.M.

[Total Marks : 100

- Instructions :** (1) Figure to the right indicates full marks.
(2) Write answer of each section in separate answer sheets.
(3) Assure suitable data if required.

Section-I

- Q.1 a) What is stack? Write Push, Pop, Peep functions of stack using array. 6
b) Suppose A and B two sorted linked list with distinct elements. Write algorithm to combine the lists into single sorted list C. 6
c) What is recursion? Explain Tower of Hanoi with the help of recursion with 3 discs. 6

- Q.2 a) Define Queue. Write algorithm or C function for all operations on queue. 6
b) Explain double linked list. Write algorithm or C function which removes the first element of a list and add it to the end of the list without changing any values. 6
c) Show the evaluation of following postfix notation using stack. 4
abc*+de/-
Take value of a=2, b=3, c=4, d=6 and e=3.

OR

- Q.2 a) Define Circular Queue. Write algorithm or C function for inserting and deleting an element in circular queue. 6
b) Write algorithm of C program for following string operations. 6
(1) In a given text T, replace the all occurrence of a pattern P1 by a Pattern P2. i.e replace(T, P1, P2)
(2) In a given text T, delete every occurrence of a pattern P. i.e. delete(T,P)
c) Give the structure representation of Polynomial equation P with 3 variables X, Y, Z and showing its power and co-efficient. 4
Suppose P1 and P2 two polynomials which are stored as a linked lists. Write algorithm or C program to add P1 and P2 in P3.

- Q.3 Attempt Following . (Any Four) 16
a) Pre-order and Post-order tree traversal.
b) Simulation and its application.
c) Dynamic memory representation and static memory representation
d) Quick sort
e) Translate $(A / B * C) * (E + F / G)$ in to post fix and Pre fix notation.

[P. T. O.